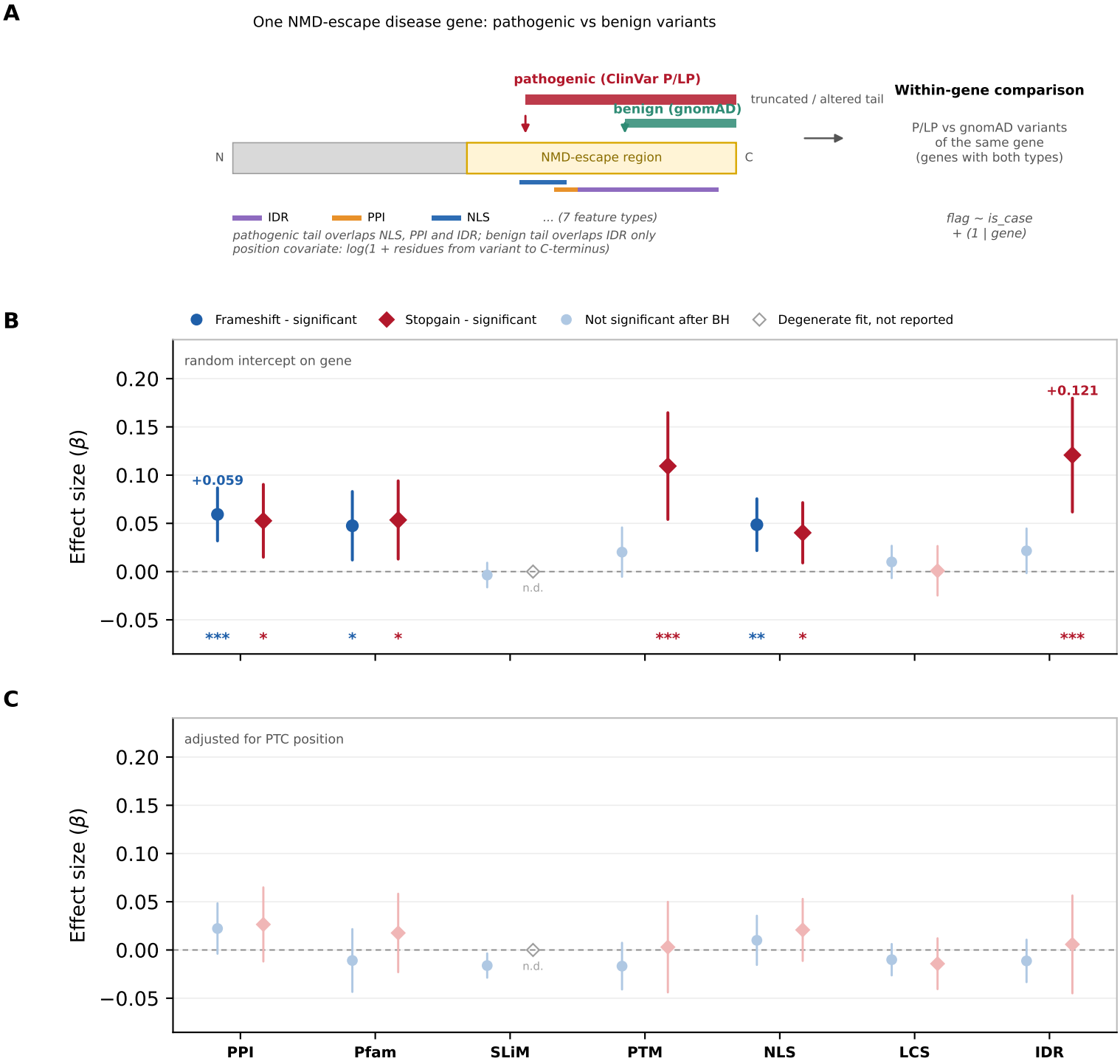
A
One NMD-escape disease gene: pathogenic vs benign variants
pathogenic (ClinVar P/LP)
truncated / altered tail
benign (gnomAD)
N
NMD-escape region
C
IDR
PPI
NLS
... (7 feature types)
pathogenic tail overlaps NLS, PPI and IDR; benign tail overlaps IDR only
position covariate: log(1 + residues from variant to C-terminus)
Within-gene comparison
P/LP vs gnomAD variants
of the same gene
(genes with both types)
flag ~ is_case
+ (1 | gene)
B
Frameshift - significant
Stopgain - significant
Not significant after BH
Degenerate fit, not reported
random intercept on gene
Effect size (β)
0.20
0.15
0.10
0.05
0.00
−0.05
+0.059
+0.121
n.d.
***
*
*
*
***
**
*
***
C
adjusted for PTC position
Effect size (β)
0.20
0.15
0.10
0.05
0.00
−0.05
n.d.
PPI
Pfam
SLiM
PTM
NLS
LCS
IDR